

Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

<think>
Distance to and from the house = 3 hours * 60 mph = 180 miles
Distance covered in the first 2 hours of the return trip = 0 mph * 2 hours = 0 miles
Distance covered in the next 30 minutes at 30 mph = 0.5 hours * 30 mph = 15 miles
Distance covered in the last 1.5 hours of the return trip = 1.5 hours * 80 mph = 120 miles
Total distance covered in the 4 hours return trip = 0 + 15 + 120 = 135 miles
Distance remaining to the house = 180 miles - 135 miles = 45 miles
</think>
<answer> 45 </answer>

